

Team Capability Profile - Adrian-Florin ANTON

Role / Title: Mechanical Engineer

Employment Type: Full time

Location / Timezone: Bucharest, EET

Start Date / Tenure: August 2023 - present

Reporting Line: Reports to Head of Engineering / CTO

Short Bio

PhD candidate in Engineering with a funded research topic in robotics, top graduate of the **MSc** program in **Advanced Mechatronics (valedictorian)**, and hands-on **Mechanical Engineer**. Passionate about robotics and real-world hardware, I combine strong theoretical background with practical end-to-end development skills, from mechanical design and prototyping to on-robot validation, driven by curiosity, rapid learning and a constant desire to explore and apply emerging technologies in dynamic engineering environments.

1) Scope of Responsibility (What you own)

- Component / subsystem owned: End-to-end mechanical architecture of autonomous cleaning robots:
 - Design of metal chassis and load-bearing structures
 - Functional and prototype parts produced via 3D printing (FDM & SLA)
 - Laser-cut plexiglass and plastic – personal operation of the CO₂ laser cutter
 - Design of parts for CNC milling and turning (DFM, supplier coordination + in-house validation)
 - Design of silicone components and dedicated molds with focus on castability; occasional execution of the casting process
 - Hands-on assembly of mechanical sub-modules and 3D-printed parts
 - Creation and maintenance of full technical documentation for all manufactured parts (CAD models, drawings, tolerances, production files)
- What “done” means for you:
 - CAD designs released and physically fabricated or validated

- Laser-cut and 3D-printed parts produced, finished and assembled by you when required
 - Sub-assemblies mounted on robots with zero blocking mechanical issues
 - Mechanical systems validated through physical prototyping, not only on drawings
- Interfaces with: Electrical Engineering, Embedded / Firmware, Software, Manufacturing & Production, QA / Field Testing, Procurement & External Suppliers

2) Key Contributions at Agora

- Owned end-to-end mechanical development of autonomous cleaning robots, from CAD design to physical fabrication and on-robot validation.
- Established rapid in-house prototyping capabilities (CO₂ laser cutting, 3D printing, silicone casting, vacuum forming), reducing hardware iteration cycles from days to hours.
- Coordinated CNC, bending and welding suppliers, ensuring manufacturability, dimensional accuracy and on-time delivery.
- Designed the complete mechanical architecture of **Sweep robots from D1 to D8 and E1** evolving the platform from early CAD-based prototypes to small-series production-ready systems, including major redesigns of chassis, water tanks and cleaning units to improve capacity, clog resistance, robustness and serviceability, while also developing the docking station from basic test rigs to IPX6-certified metal versions with reliable water loading, detergent dosing and waste evacuation, and creating in-house silicone components and molds that reduced part costs by more than 5x compared to external suppliers.

3) Core Competencies

Robotics / Mechatronics:

- Mechanical architecture for mobile robots
- Robot–environment interaction mechanisms
- Integration of mechanical, electrical subsystems

Manufacturing & Prototyping:

- 3D printing (FDM, SLA), including setup, calibration and post-processing.
- CO₂ laser cutting – setup, calibration, material selection
- Design for CNC & lathe manufacturing (DFM knowledge)
- Silicone mold design & casting, gravity and injection molding

- Thermoforming & vacuum forming processes

Software / Tools:

- Fusion 360, AutoCAD,
- G-code basics, CAD documentation

Field / Hands-On:

- Mechanical sub-assembly, finishing, fitment corrections
- Dimensional inspection, tolerance validation
- Rapid iteration cycles between CAD design and real hardware

4) Deliverables in Next 90 Days (Forward-looking)

- Q1 deliverables owned:
 - Release of Chassis vNext with improved rigidity and simplified assembly flow
 - Library of standardized 3D-printed functional parts for faster prototyping
 - Manufacturing documentation pack for serial production
- Risks / dependencies: supplier lead-times for CNC and bent metal parts, process stability for thermoforming and silicone molding, dimensional deviations caused by welding processes on parts requiring tight tolerances
- Definition of success: $\geq 30\%$ reduction in mechanical rework during prototype assembly. First-pass mechanical assembly success rate $\geq 95\%$. Extended maintenance intervals with significantly fewer field interventions on deployed robots